INCREMENTALITY STUDY · PORTFOLIO PROJECT 2

Did the Ads Actually Work?

Measuring what retargeting really causes, who it persuades, whether it pays, and how to measure it next time

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Independent analysis of public data: Criteo Uplift v2.1 (CC BY-NC-SA 4.0) and Iowa Liquor Sales. Not affiliated with or endorsed by Criteo, Google or the State of Iowa.

Executive summary

A retailer pays for retargeting: ads shown to people who have already visited its website. Its dashboards credit those ads with a large share of sales. The question is simple: **how many of those sales would have happened anyway?**

I answered it with a real randomised ad test run by Criteo on 13,979,592 users: 85% were allowed to see the ads and 15% had them blocked, so any difference between the groups was caused by the ads.

What I found

- **Only 54 of every 100 conversions credited to ads were caused by them** (95% interval 50 to 58). The other 46 would have happened anyway. Dashboards built on attribution overstate retargeting's return by about 1.9x.
- **The dashboard view is wildly misleading.** Users who saw an ad converted 41x more often than users who didn't. But they were already 19x likelier to buy; the real effect of seeing an ad is 2.2x.
- **The effect is concentrated.** About 91% of it sits in the top 20% of users, and half of today's spend, aimed at likely buyers, keeps 84% of the conversions ads cause. A simple likely-buyer model finds these users as well as more advanced uplift models.
- **It pays, but only in part.** On illustrative numbers (dashboard ROAS 8, margin 30%), each £1 of ads looks like +£1.40 of profit but is really +£0.29. The best half of spend makes +£1.16 per £1; the other half loses about 58p. With a 30% margin, a campaign needs a dashboard ROAS above 6.2 just to break even.
- **When users can't be held out, regional tests are blunt.** On real Iowa sales with 2,700 simulated campaigns, even 20 counties for 12 weeks only reliably detected a lift of about 9%, and a CausalImpact-style model raised false alarms 10% to 18% of the time instead of 5%.

What to do

1. Report incremental ROAS (dashboard ROAS x caused share) next to attributed ROAS in every campaign review.
2. Keep an always-on holdout, sized properly: rare conversions need scale (about 4.9 million users to see a +10% conversion lift with a 10% holdout).
3. Test cutting the weaker half of spend with its own holdout before cutting it.
4. Use geo tests only where holdouts are impossible, sized by simulation and placebo-checked first.

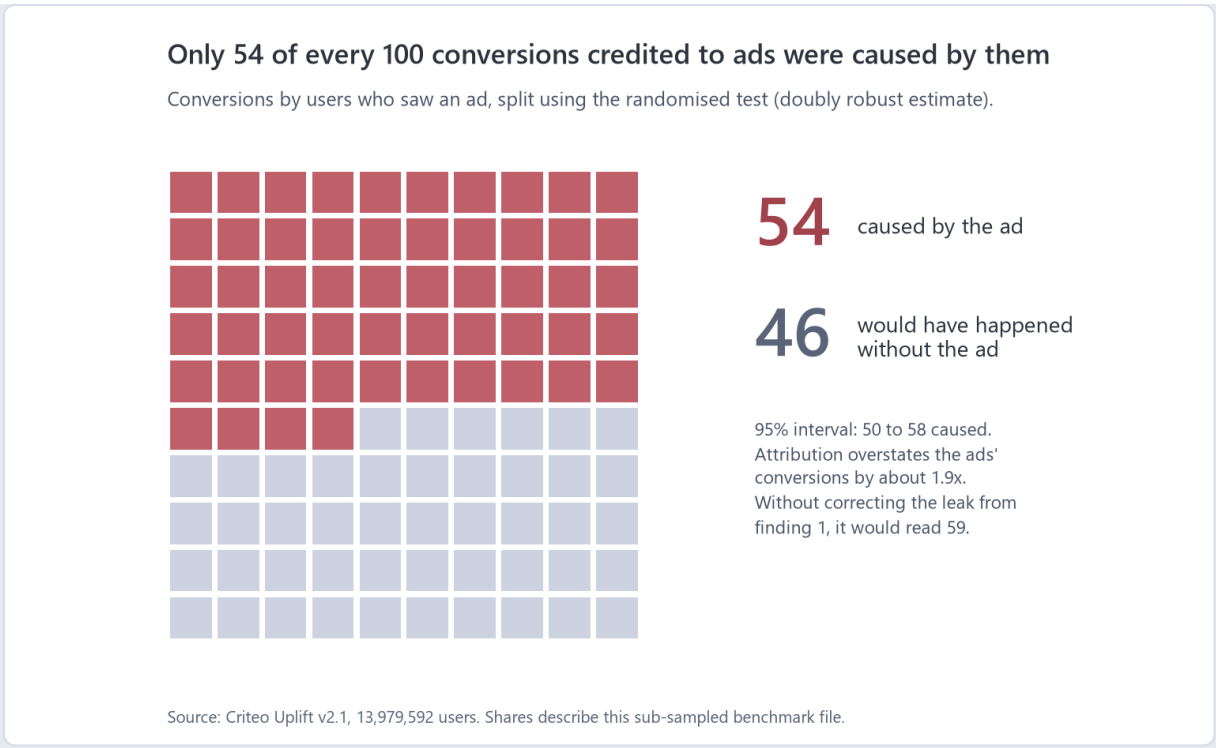

Only 54 of every 100 conversions credited to ads were caused by them
Conversions by users who saw an ad, split using the randomised test (doubly robust estimate).
54
caused by the ad
46
would have happened without the ad
95% interval: 50 to 58 caused.
Attribution overstates the ads' conversions by about 1.9x.
Without correcting the leak from finding 1, it would read 59.
Source: Criteo Uplift v2.1, 13,979,592 users. Shares describe this sub-sampled benchmark file.

Figure 1

1. The question and the data

Why attribution can't answer it

Attribution hands credit for a sale to the ads a person saw before buying. It cannot say whether the person would have bought anyway, and retargeting aims precisely at people who are already likely to buy. My previous project, a review of Google's Merchandise Store, showed attribution models disagree with each other and that its marketing mix model could not be calibrated without experiments. This project supplies the experiment.

The Criteo test (main study)

Criteo, an ad-tech company, ran randomised incrementality tests and released 13,979,592 users from them. For each user: which group they were in (85% treatment, 15% control), whether they actually saw an ad, whether they visited the advertiser's site and whether they converted, plus 12 anonymised features recorded before any ad was shown.

Only 3.6% of the treatment group actually saw an ad: being allowed to see ads doesn't mean you will. No control user saw one. That is what makes the effect of seeing an ad measurable.

Limits stated up front. Criteo sub-sampled the file so that its real incrementality can't be recovered, so the absolute numbers describe this benchmark file, not Criteo's live campaigns. There are no prices or costs, so money figures use stated assumptions. The features are anonymised, so persuadable users can be described by behaviour but not by who they are.

Iowa liquor sales (methods demonstration)

To grade regional testing methods I needed real sales where the true campaign effect is known. I pulled weekly sales for Iowa's counties from BigQuery (2016 to 2026; 96 counties after leaving out three with many empty weeks), then added campaigns of known size myself. Every chart from this part is labelled "real data, simulated campaign".

2. Is the test sound? (Finding 1)

Before trusting any lift number I checked the randomisation three ways.

- **The split** came out at exactly 85.00% treated, as designed.
- **Balance:** on every feature, treated and control users differ by less than 0.049 standard deviations, well inside the usual 0.1 line.
- **A classifier test:** if assignment were random, no model could tell the groups apart from the features. On 1 million users a model scored an AUC of 0.509, where 0.5 is a coin flip. A placebo with shuffled labels scored 0.499, so the small edge is real.

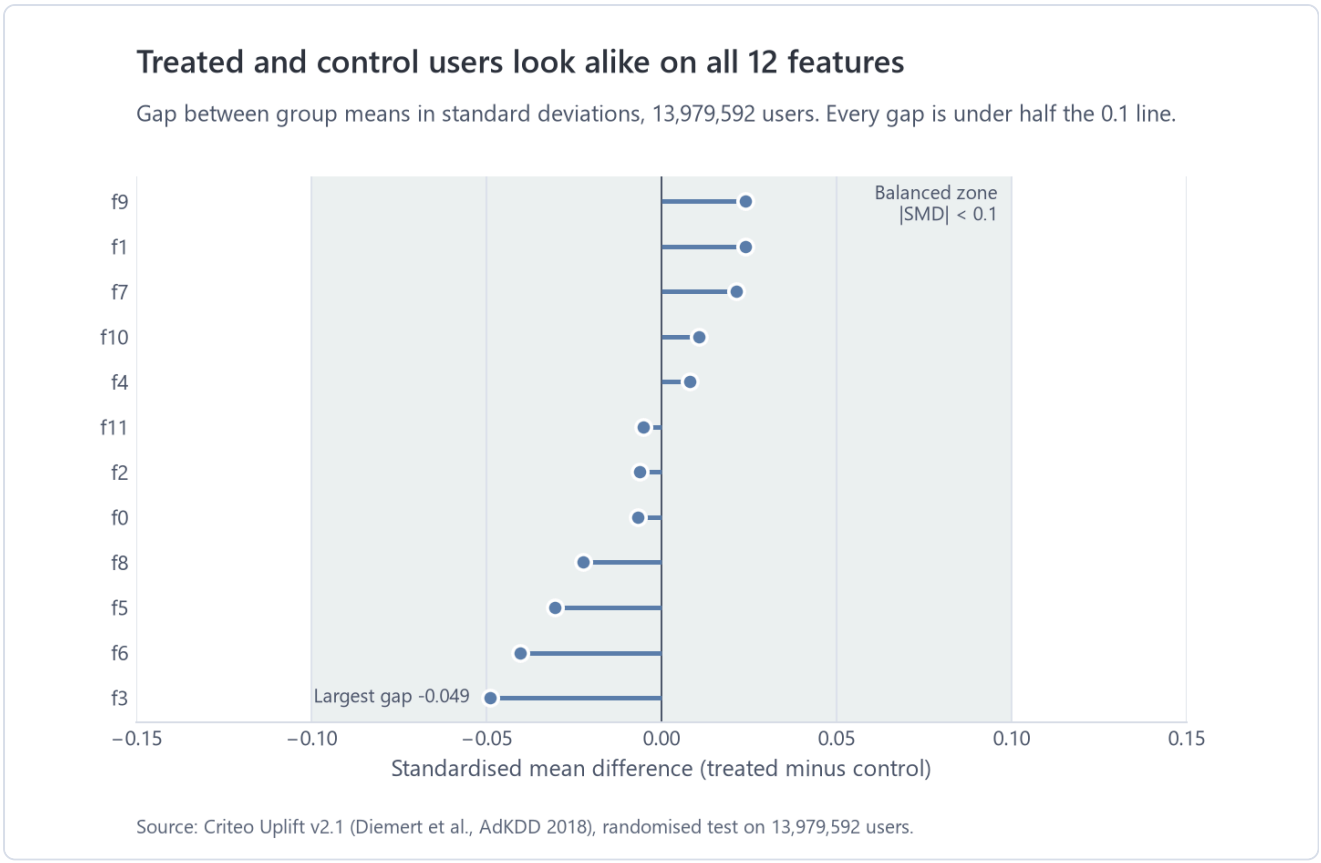


Figure 2

Criteo's own paper explains why: the file merges several advertisers' tests that had different treated shares. The first release leaked treatment information because of it; version 2 tried to fix it, and Criteo's check found no leak. At 14 million users, a small one remains.

The features were recorded at the moment users were assigned, before any ad, so adjusting for them is valid. That adjustment matters for the next finding.

3. Dashboards vs reality (Findings 2 to 4)

The dashboard view

A dashboard compares users who saw an ad with users who didn't: 5.38% against 0.13%, which looks like a 41x boost.

The real lift for the whole targeted group

Comparing the randomly assigned groups as a whole, the treatment group converted **52% more** than control and visited **20% more**. The raw comparison says +59% and +27%; the small leak from Finding 1 inflated it. Two different adjustments (a linear one and a doubly robust one with cross-fitting) agree on the corrected figures.

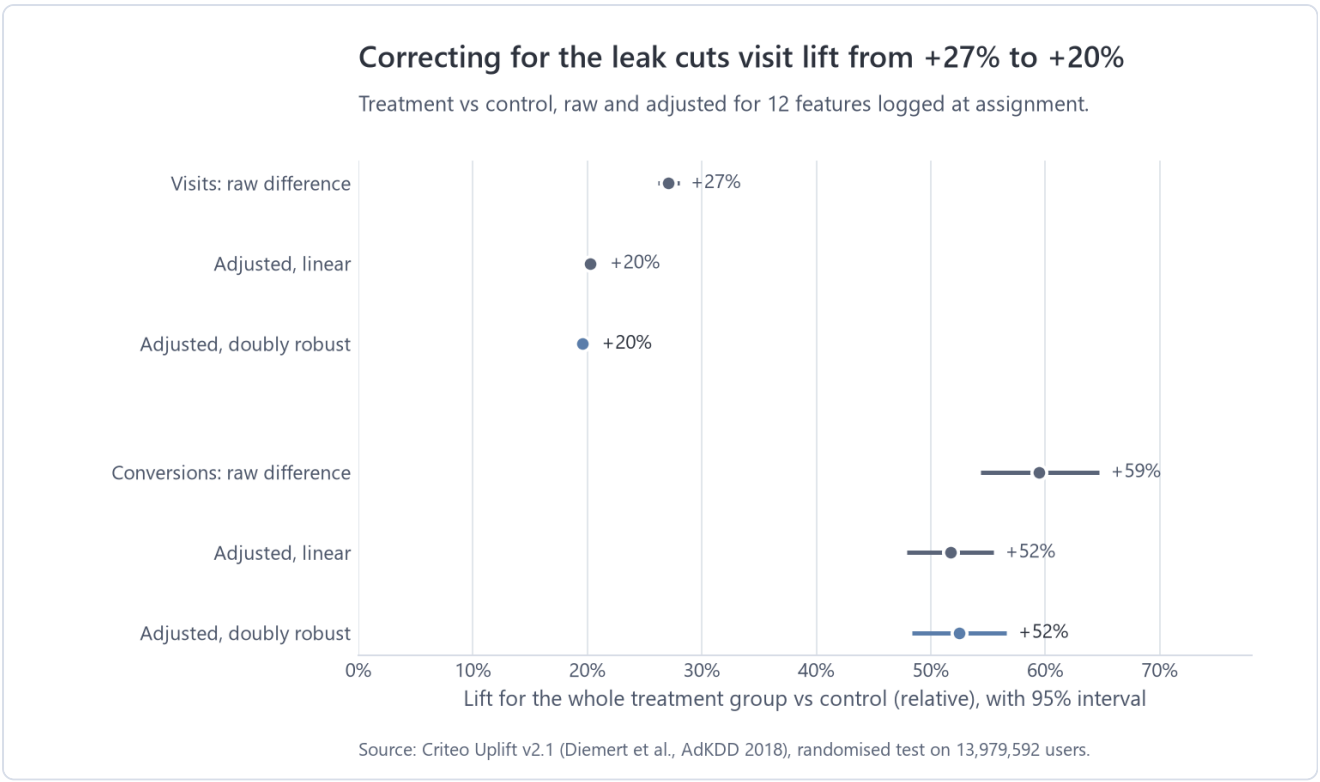


Figure 3

The effect of actually seeing an ad

Because nobody in control could see an ad, the random assignment works as an instrument: the whole-group lift divided by the share who saw an ad gives the effect on those who did (two-stage least squares gives the same number). Users who saw an ad converted at 5.38%; without it they would have converted at about 2.49%, already 19x the rate of everyone else. The ad's real effect is **+2.89 points, or 2.2x**, not 41x.

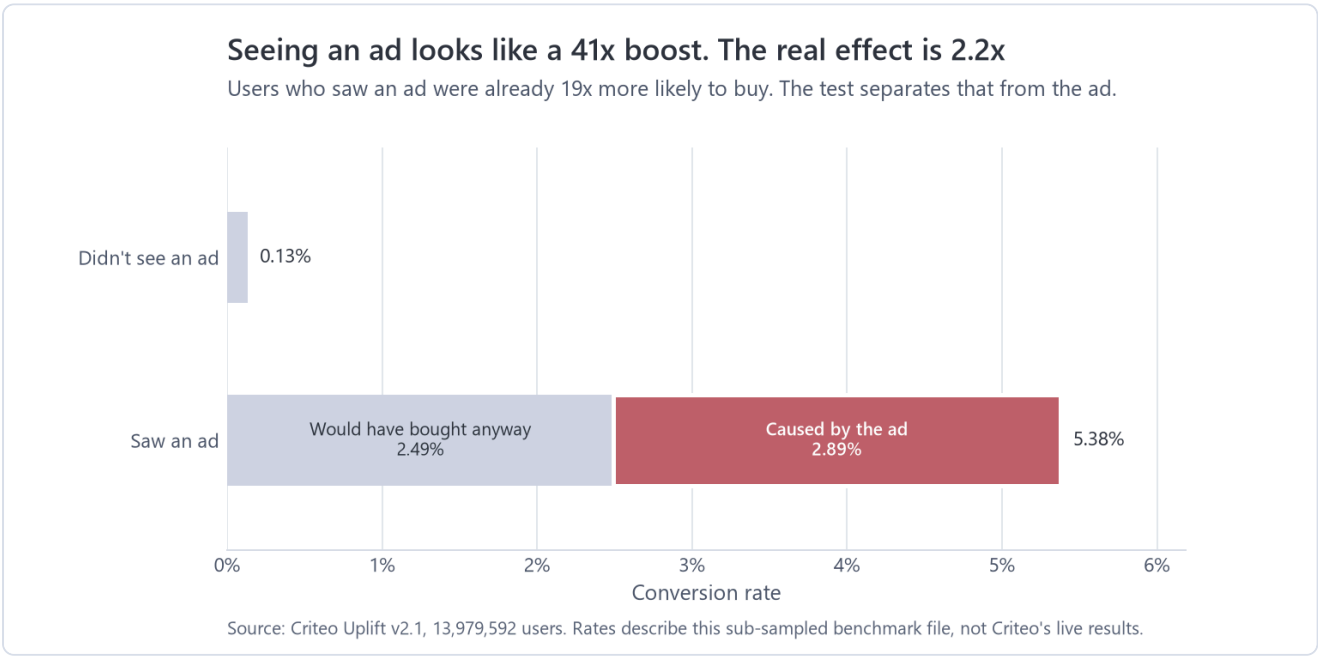


Figure 4

4. Credited vs caused (Finding 5)

Attribution based on ad views credits every conversion by a user who saw an ad. The test shows that **only 54 of every 100 of those were caused by the ad** (95% interval 50 to 58). Without the correction from Finding 1, the figure would read 59.

So an attribution report overstates retargeting's conversions, and its return on spend, by about **1.9x**. Ads do work: seeing one roughly doubles a user's chance of buying. The problem is that attribution also hands them the sales of keen shoppers who were going to buy anyway.

5. Who is persuaded? (Finding 6)

Some users buy whether or not they see an ad; only persuadable users are worth paying to reach. I trained uplift models (a T-learner and an X-learner) on 80% of users to predict each user's response to being targeted, and graded them on the other 2.8 million against a simple "likely buyers" model, the kind retargeting usually uses.

- **The effect is concentrated:** the top 20% of users hold about **91%** of it, for visits and for conversions.
- **The simple model is as good as the advanced ones** for conversions (a cut-off fixed before seeing results). Both pick largely the same people, because only users who are shown ads can be persuaded and ads go to keen shoppers. The uplift model wins only for visits on a tight budget (top 10% of users: 80% vs 68% of the effect).
- **Measured by spend**, the yardstick that matters for money: half of today's ad spend, aimed at likely buyers, keeps **84%** of the conversions the ads cause (78% to 90%) and 69% of the visits.

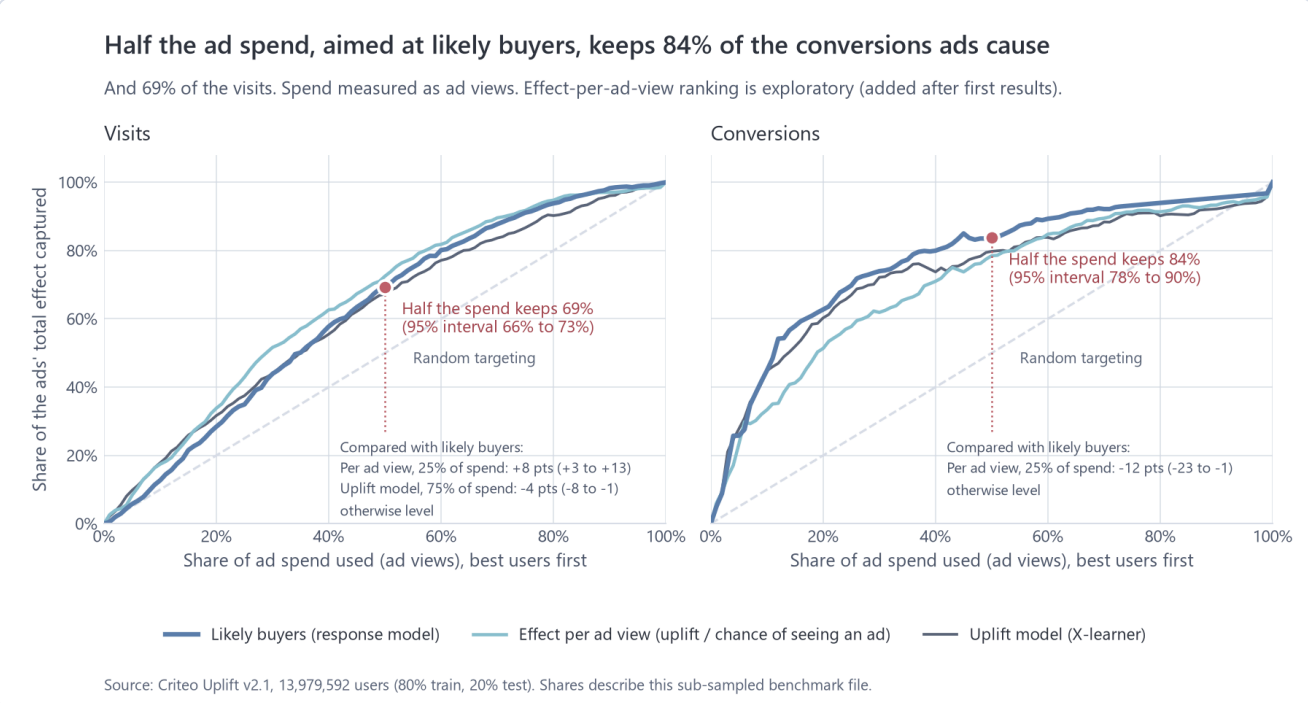


Figure 5

Two traps were avoided. Grading by head count alone would have overstated the savings, because the top users already receive most of the ad views. And the conversion models gave large groups of users identical scores; a plain sort would have ordered them by their position in the file, which is linked to treatment, so tied users are counted fairly.

6. Is it worth the money? (Finding 7)

The data has no prices or costs, so I used two assumptions anyone can swap: a dashboard ROAS of 8 and a 30% profit margin. Everything else comes from the test.

Profit per £1 of ad spend	Estimate (95% interval)
What the dashboard implies	+£1.40
Real, all spend	+£0.29 (+£0.19 to +£0.38)
Real, best half of spend	+£1.16 (+£0.94 to +£1.39)
Real, other half of spend	-£0.58 (-£0.74 to -£0.42)

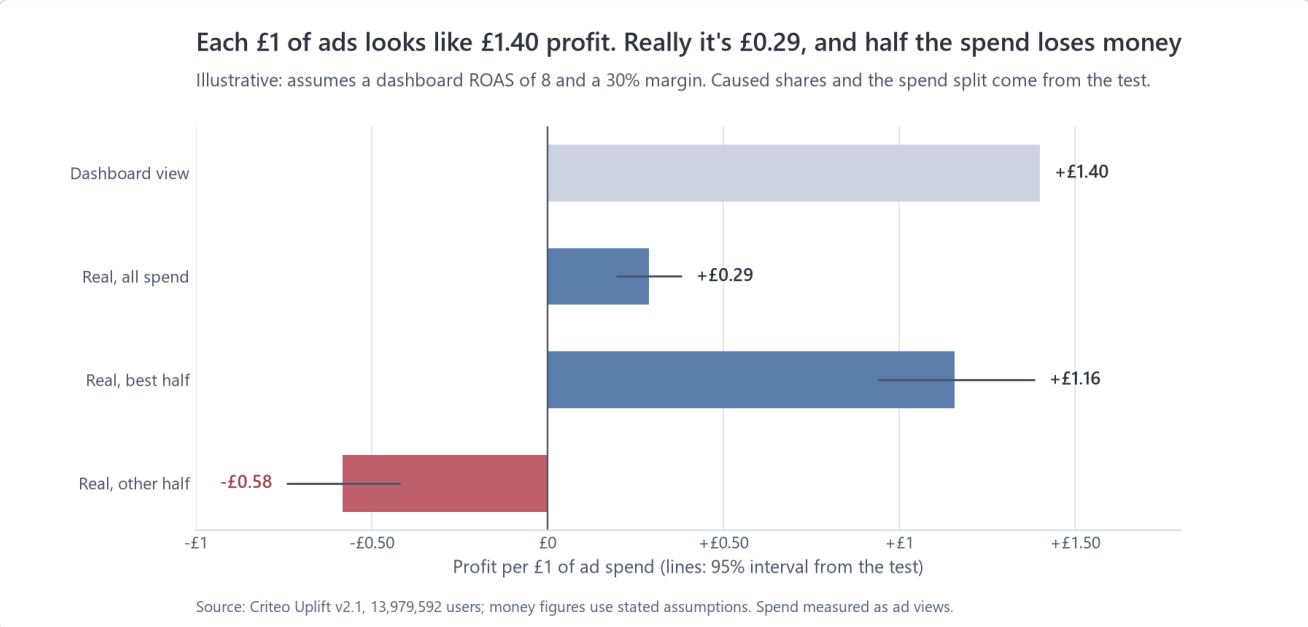


Figure 6

Cutting the weaker half of spend would roughly **double profit** (2.0x) while giving up 16% of the conversions the ads cause. A rule that needs no ROAS assumption: **with a 30% margin, a campaign needs a dashboard ROAS above 6.2 to break even, and its weaker half needs 19.2.** Across dashboard ROAS of 4 to 15 and margins of 20% to 50%, the weaker half loses money in 21 of 24 combinations.

7. When you can't hold users out: geo tests (Finding 8)

TV, outdoor and some ad platforms can't hold out individual users. The fallback is a geo test: run the campaign in some regions, not others. I graded three methods on 2,700 simulated campaigns in real Iowa county sales.

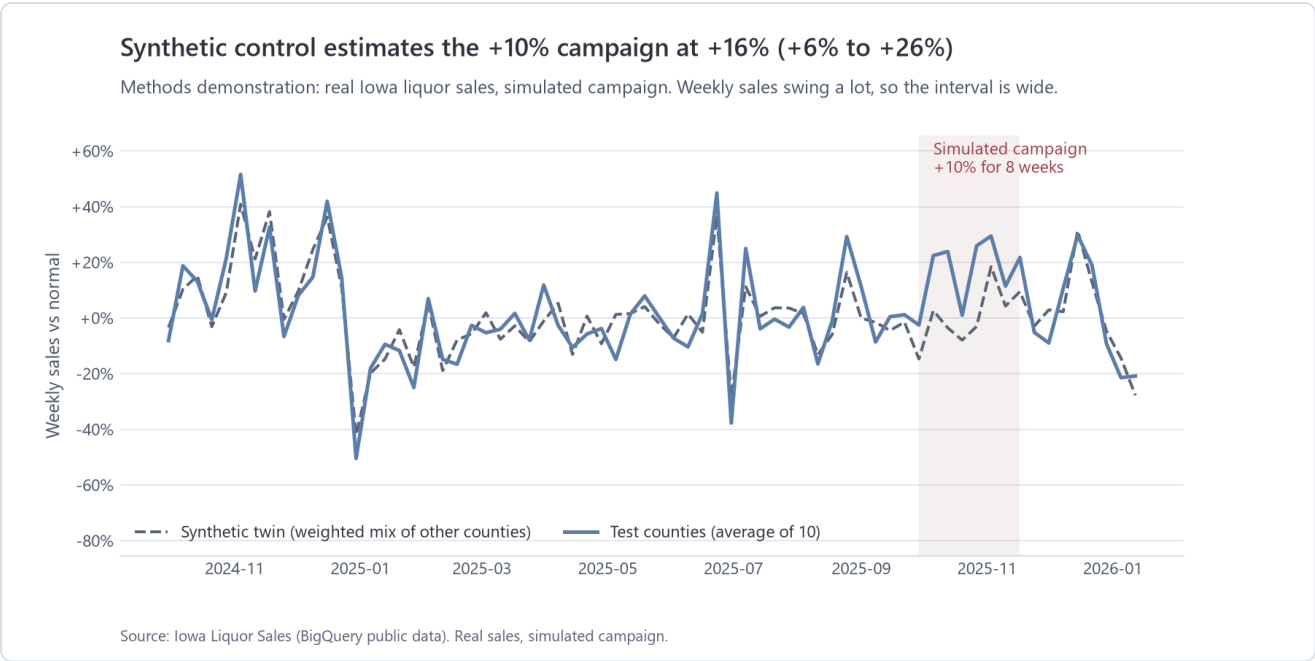


Figure 7

- **Two simple methods stay honest.** Difference-in-differences and synthetic control raised false alarms 3% to 8% and 4% to 7% of the time, close to the promised 5%.
- **The CausalImpact-style model cries wolf** 10% to 18% of the time, 2 to 3.5 times too often, and was the least accurate.

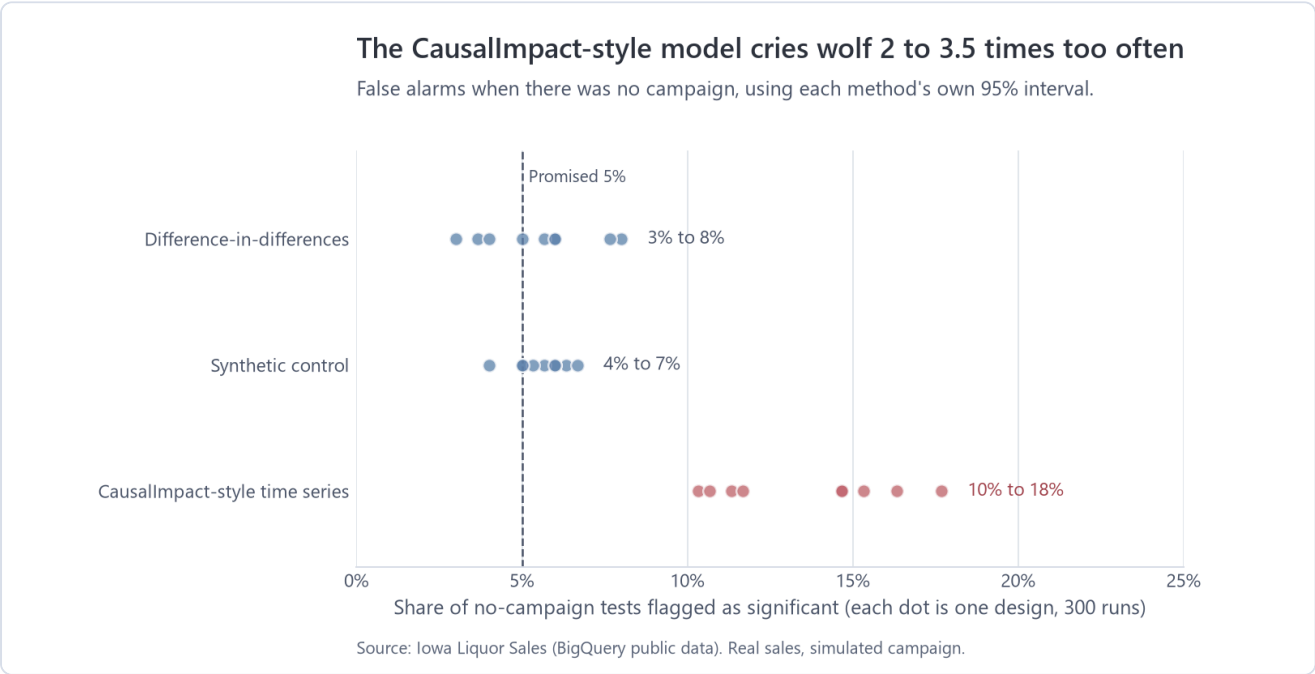


Figure 8

- **Only big effects are detectable.** Weekly county sales swing by about 50% from week to week. Even 20 counties for 12 weeks only reliably detect a lift of about 9%; 5 counties for 4 weeks need over 20%. Typical ad effects of a few percent would usually be missed.

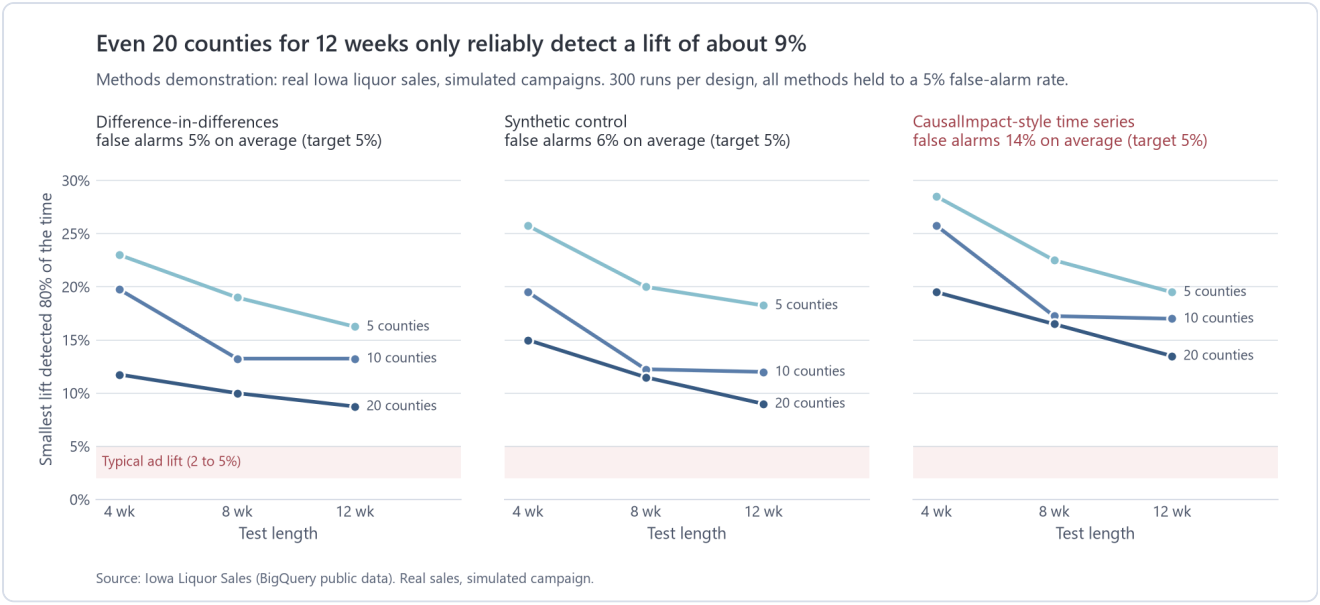


Figure 9

8. Recommendation: a test-and-learn measurement plan (Finding 9)

1. **Judge retargeting on what it causes.** Real ROAS = dashboard ROAS x caused share.
2. **Keep an always-on holdout, sized properly.** I built a sizing calculator from the test's own rates and checked it against the precision the Criteo test achieved (within 0.2 points). Conversions are rare, so a 10% holdout needs about **4.9 million users** to see a +10% conversion lift (1.8 million with a 50/50 split). Visits give an early read with about 0.9 million.

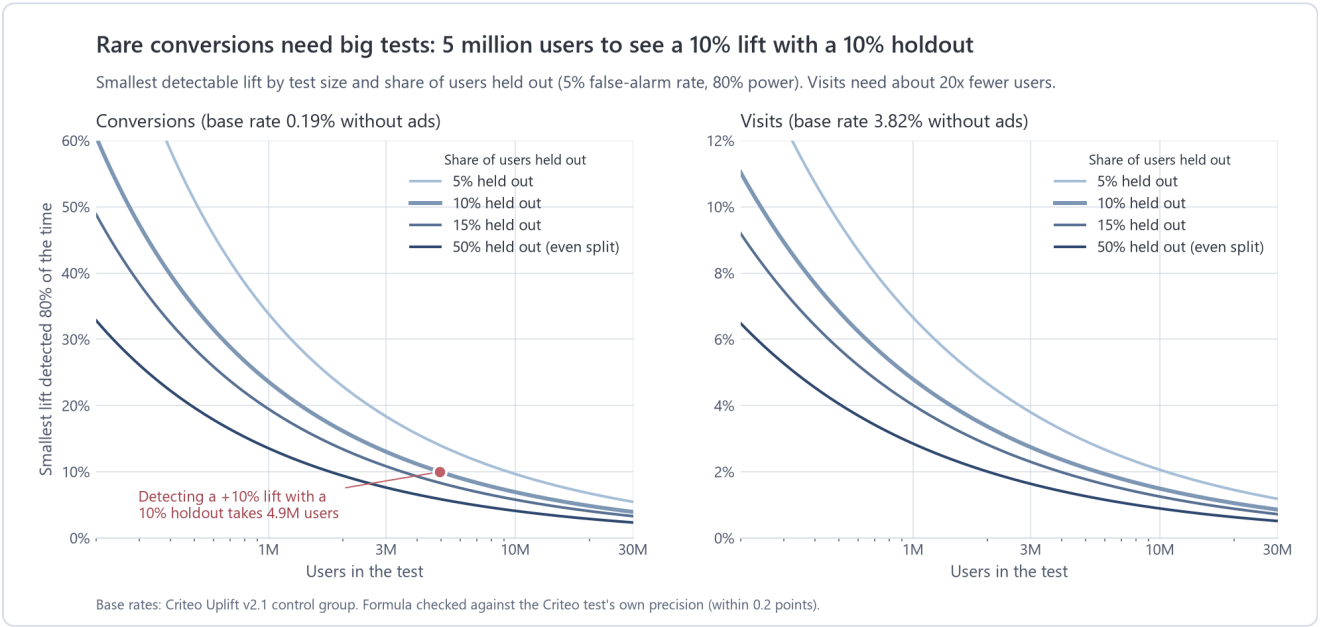


Figure 10

3. **Test the budget cut with its own holdout** before making it.
4. **Start with the simple likely-buyer model**; revisit uplift modelling with more data.

5. **Health-check every experiment** (split, balance, classifier with placebo) and adjust for pre-treatment features.
6. **Use geo tests only where holdouts are impossible**, with difference-in-differences or synthetic control, sized by simulation and placebo-checked.
7. **Use a marketing mix model for cross-channel budgets, anchored by these experiments.**

Question	Best method	Watch out for
Is this digital campaign working?	User-level holdout	Holdout too small for rare conversions
Who should we target?	Likely-buyer or uplift model, checked by holdout	Grade by spend, not head count
Is a channel with no user-level control working?	Geo test (DiD or synthetic control)	Detects only large effects on noisy data
How should we split budget across channels?	MMM calibrated with experiments	Correlation unless anchored by tests
Should we cut part of the budget?	Holdout on that part	Needs enough users in that part

Before trusting any ad number, ask: is it attributed or incremental? Was the comparison randomised and health-checked? Could a test this size see an effect this big? What is the 95% interval? Has the method been placebo-tested?

9. Limits

- **Benchmark data.** Absolute numbers describe Criteo's sub-sampled file; every business must measure its own caused share. The methods transfer.
- **Money assumptions.** ROAS and margin are illustrative; the break-even rule and sensitivity tables show how the conclusions move.
- **Spend is proxied by ad views.** Real cost also depends on how many ads each user sees and their price.
- **One geo dataset.** Liquor wholesale orders are lumpy; smoother sales give smaller detectable lifts.

Appendix A: methods in brief

Step	Method	Code
Experiment health	Chi-square sample ratio test; standardised mean differences; gradient-boosted classifier two-sample test with placebo	`experiment.py`
Whole-group lift	Difference in proportions with 95% intervals; Lin (2013) regression adjustment; doubly robust AIPW with 2-fold cross-fitting	`lift.py`
Effect of seeing an ad	Wald / two-stage least squares (linearmodels), robust errors	`lift.py`
Uplift	T- and X-learner (gradient boosting), graded on held-out users with doubly robust pseudo-outcomes, paired intervals, tie-aware	`uplift.py`
Money	Incremental ROAS and profit per £, uncertainty by simulation	`money.py`
Geo tests	Collapsed difference-in-differences (Bertrand, Duflo and Mullainathan 2004); synthetic control with placebo inference; structural time series (statsmodels)	`geo.py`
Sizing	Minimum detectable relative lift on the log-ratio scale	`planning.py`

Appendix B: reproduce

Everything is rebuilt by `python python/scripts/run_all.py`; every number in this report is printed by `python python/scripts/headline_facts.py`. Code, notebooks and write-ups per finding are in the repository.

Appendix C: glossary

- **Incrementality:** sales caused by the ads, beyond what would have happened anyway.
- **Attribution:** rules that hand out credit for a sale to the ads a person saw before buying.
- **Intent-to-treat:** treatment group minus control group, as randomly assigned.
- **Instrumental variable:** something random that changes who gets treated, used to isolate the treatment's effect.
- **Doubly robust:** an estimate that stays right if either its model of who gets treated or its outcome model is right.
- **Uplift model:** a model that predicts how much an ad changes each person's chance of buying.
- **Geo test:** an experiment in which regions, not people, are the test and control groups.
- **Minimum detectable lift:** the smallest real effect a test catches 80% of the time.